A literature comparison matrix

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete study matrix and comparison-gap record.

Keep source-specific question, unit, metric, data, conditions, unknowns and next evidence together.

Keep one habit: Recall → W1–W3/W5 → independent metric mismatch → literature-matrix project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Replication and reproduction. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Finding primary research. Give an example and a mistake that the idea helps you avoid.

Replication and reproduction

> **My remembered idea, example and correction**

Finding primary research

> **My remembered idea, example and correction**

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A literature matrix compares studies by aligned meaning, not headline-score sorting. Preserve question/task, data/population, unit, baseline/model version, metric/cutoff/aggregation, split, resources, source locator and limitations. Main fictional A reports 90% classification accuracy on described-easy cases; B 80% claim support for described-difficult generated answers.

  Tasks/metrics differ; raw denominators, versions and many conditions are not reported. Do not invent 9/10 or8/10 or assume two unknown fields match. Separate picture S1 indoors 8/10=80%, S2 outdoors 16/20=80%; same rate does not establish equivalent tools, uncertainty or tasks. Neither evaluates both tools on the same supplied photos.

  Removing indoor/outdoor column hides context without removing the real condition difference. A future common comparison needs a shared application task, eligible judged data, scoring/aggregation, versions and settings. Do not pool incompatible percentages or infer a winner from matrix completion. These are supplied fictional records, not actual literature findings, meta-analysis or performed tool evaluation.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

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- Study A reports accuracy 90% on easy classification; B reports claim support **80%** on difficult generated answers. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A literature matrix compares studies by aligned meaning, not headline-score sorting. Preserve question/task, data/population, unit, baseline/model version, metric/cutoff/aggregation, split, resources, source locator and limitations. Main fictional A reports 90% classification accuracy on described-easy cases; B 80% claim support for described-difficult generated answers.

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  Removing indoor/outdoor column hides context without removing the real condition difference. A future common comparison needs a shared application task, eligible judged data, scoring/aggregation, versions and settings. Do not pool incompatible percentages or infer a winner from matrix completion. These are supplied fictional records, not actual literature findings, meta-analysis or performed tool evaluation.

- A report omits this qualification: A matrix documents comparison limits; it does not create missing common **evidence**. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**
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MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A literature matrix compares studies by aligned meaning, not headline-score sorting. Preserve question/task, data/population, unit, baseline/model version, metric/cutoff/aggregation, split, resources, source locator and limitations. Main fictional A reports 90% classification accuracy on described-easy cases; B 80% claim support for described-difficult generated answers.

  Tasks/metrics differ; raw denominators, versions and many conditions are not reported. Do not invent 9/10 or8/10 or assume two unknown fields match. Separate picture S1 indoors 8/10=80%, S2 outdoors 16/20=80%; same rate does not establish equivalent tools, uncertainty or tasks. Neither evaluates both tools on the same supplied photos.

  Removing indoor/outdoor column hides context without removing the real condition difference. A future common comparison needs a shared application task, eligible judged data, scoring/aggregation, versions and settings. Do not pool incompatible percentages or infer a winner from matrix completion. These are supplied fictional records, not actual literature findings, meta-analysis or performed tool evaluation.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A has recall@5 .8; B has precision@3 .9 on another corpus. Can you rank them?
- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled a literature comparison matrix **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of a literature comparison matrix.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**
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MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for A literature comparison matrix. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for A literature comparison matrix. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A has recall@5 .8; B has precision@3 .9 on another corpus. Can you rank them?

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A **literature matrix** that preserves differences. Two fictional studies evaluate related tools under different conditions.
- Study S1: 8/10 correct on indoor photos. S2: 16/20 on outdoor photos. Both report **80%**.
- Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the **literature matrix** preserve?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

AI ACADEMY / AI-7/6

Same rate, different study conditions

	Correct	Total	Condition
Study S1	8	10	Indoor
Study S2	16	20	Outdoor

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/6

A literature comparison matrix

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Record each method

Keep population, task, model version and scoring in separate columns.



2. Compare rates carefully

Both rates are 80%, with different denominators and conditions.



3. Name missing comparison

Neither study directly compares both tools on the same photos.

CHANGE ONE THING

Remove the indoor/outdoor column.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A **literature matrix** that preserves differences. Two fictional studies evaluate related tools under different conditions.
- Which labelled step matches this detail? Neither study directly compares both tools on the same photos.
- 1. Name missing comparison
- 2. Record each method
- 3. Compare rates carefully
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Name missing comparison
Neither study directly compares both tools on the same photos. A **literature matrix** helps compare **evidence** without flattening differences.

Equal percentages do not mean equivalent tasks, samples or methods.

Changed-case question and optional hint

- Changed-case practice: Remove the indoor/outdoor column. Which conclusion is supported by the supplied facts?
- 1. The matrix conceals an important condition difference.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The matrix conceals an important condition difference. A reader might now mistake unlike studies for a controlled head-to-head comparison.

Repair and retain the record

Review Replication and reproduction (ai-7/5). Goal: Explain the **mechanism** and terminology of replication and reproduction. Short practice: Calculate the scores from the original and new four-record sets.

Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?. Return to this week and explain what you changed.

Review Finding primary research (ai-7/3). Goal: Explain the **mechanism** and terminology of finding primary research. Short practice: Separate what the blog claims from what the original report records.

Which methods and denominator must be inspected, and what status is honest if the original report is unavailable?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

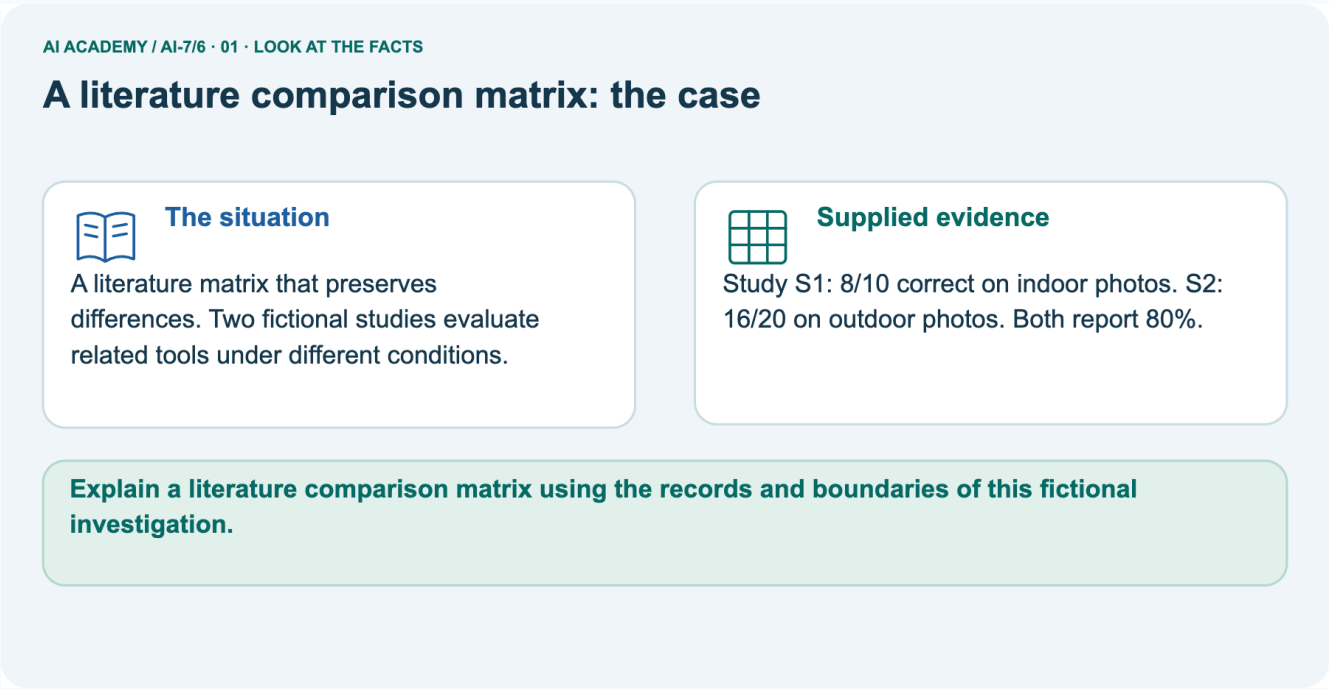
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. A literature comparison matrix: the case



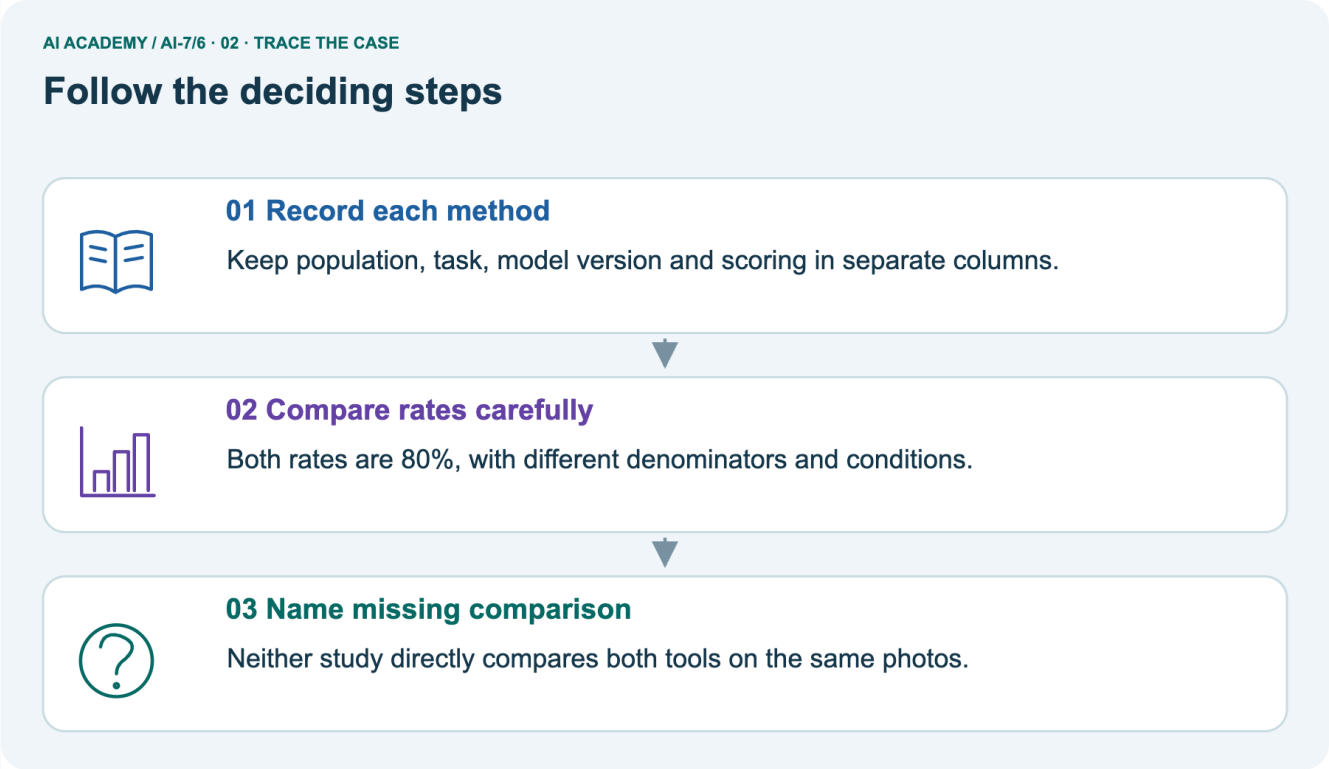
Follow the labelled picture with the matching explanation. Open the vector diagram

A **literature matrix** that preserves differences. Two fictional studies evaluate related tools under different conditions. Supplied facts: Study S1: 8/10 correct on indoor photos. S2: 16/20 on outdoor photos. Both report **80%**.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



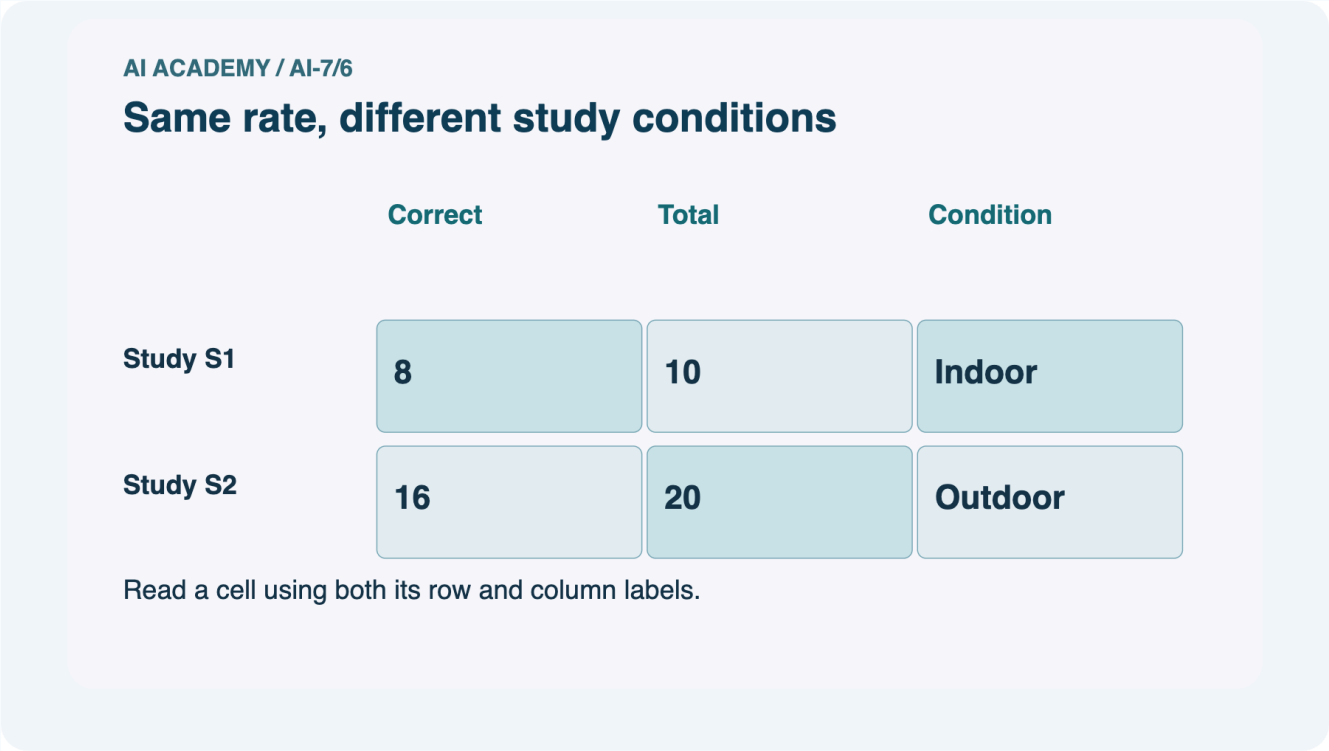
Follow the labelled picture with the matching explanation. Open the vector diagram

Record each method: Keep population, task, model version and scoring in separate columns.
→ Compare rates carefully: Both rates are **80%**, with different denominators and conditions.
→ Name missing comparison: Neither study directly compares both tools on the same photos.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Same rate, different study conditions



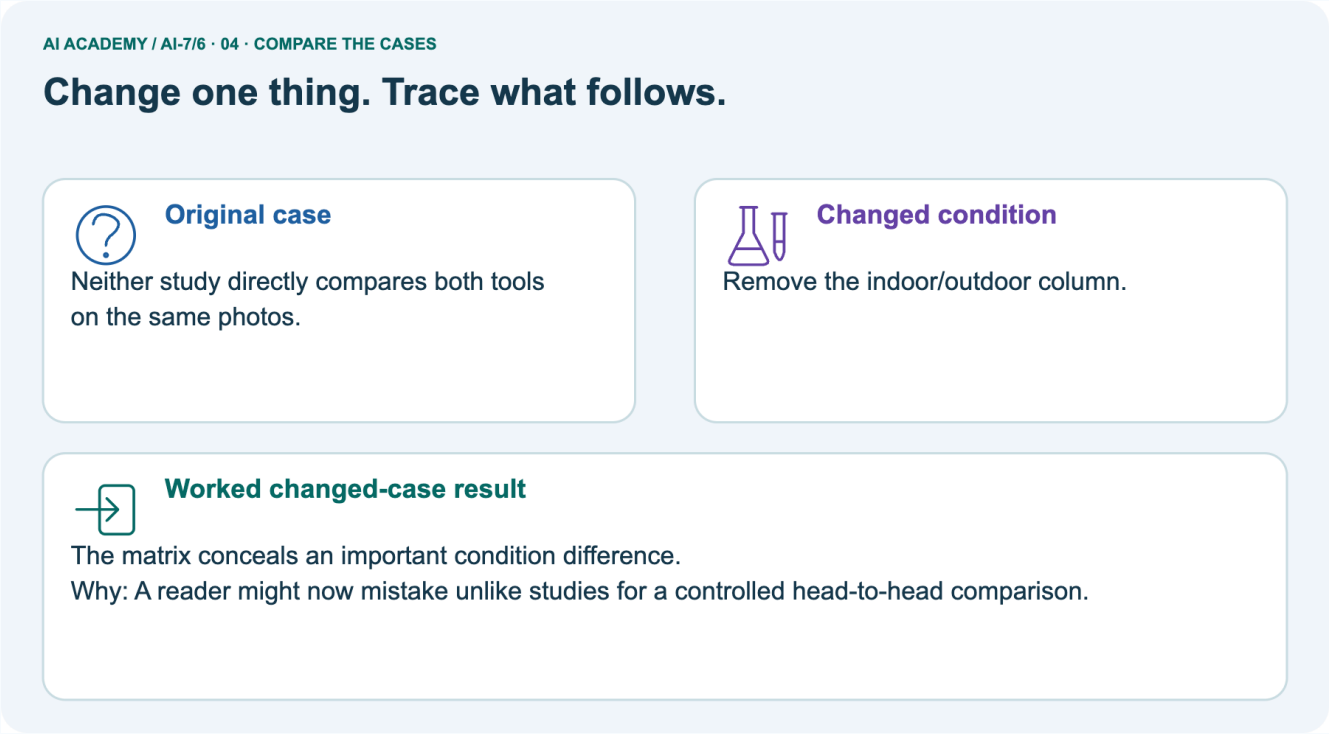
Follow the labelled picture with the matching explanation. Open the vector diagram

Same rate, different study conditions. Study S1: Correct = 8, Total = 10, Condition = Indoor. Study S2: Correct = 16, Total = 20, Condition = Outdoor.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

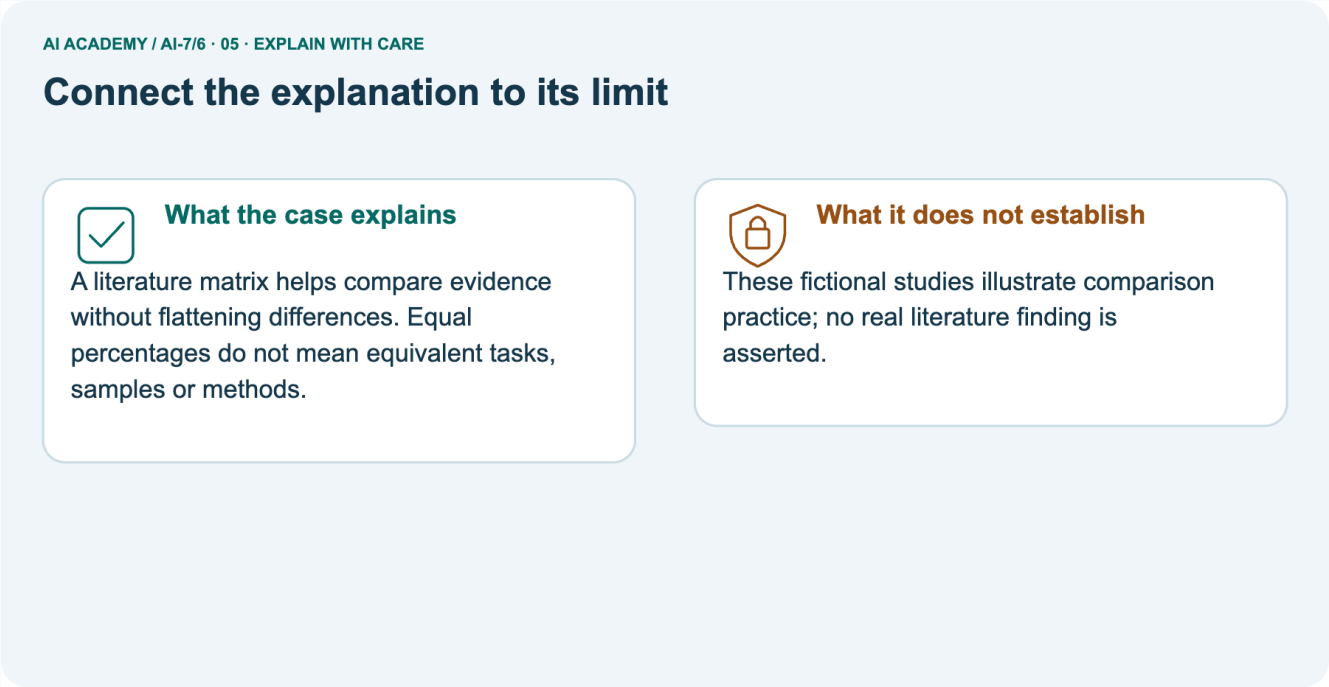
Original case: Neither study directly compares both tools on the same photos. Changed condition: Remove the indoor/outdoor column. Expected result: The matrix conceals an important condition difference.

Reason: A reader might now mistake unlike studies for a controlled head-to-head comparison.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

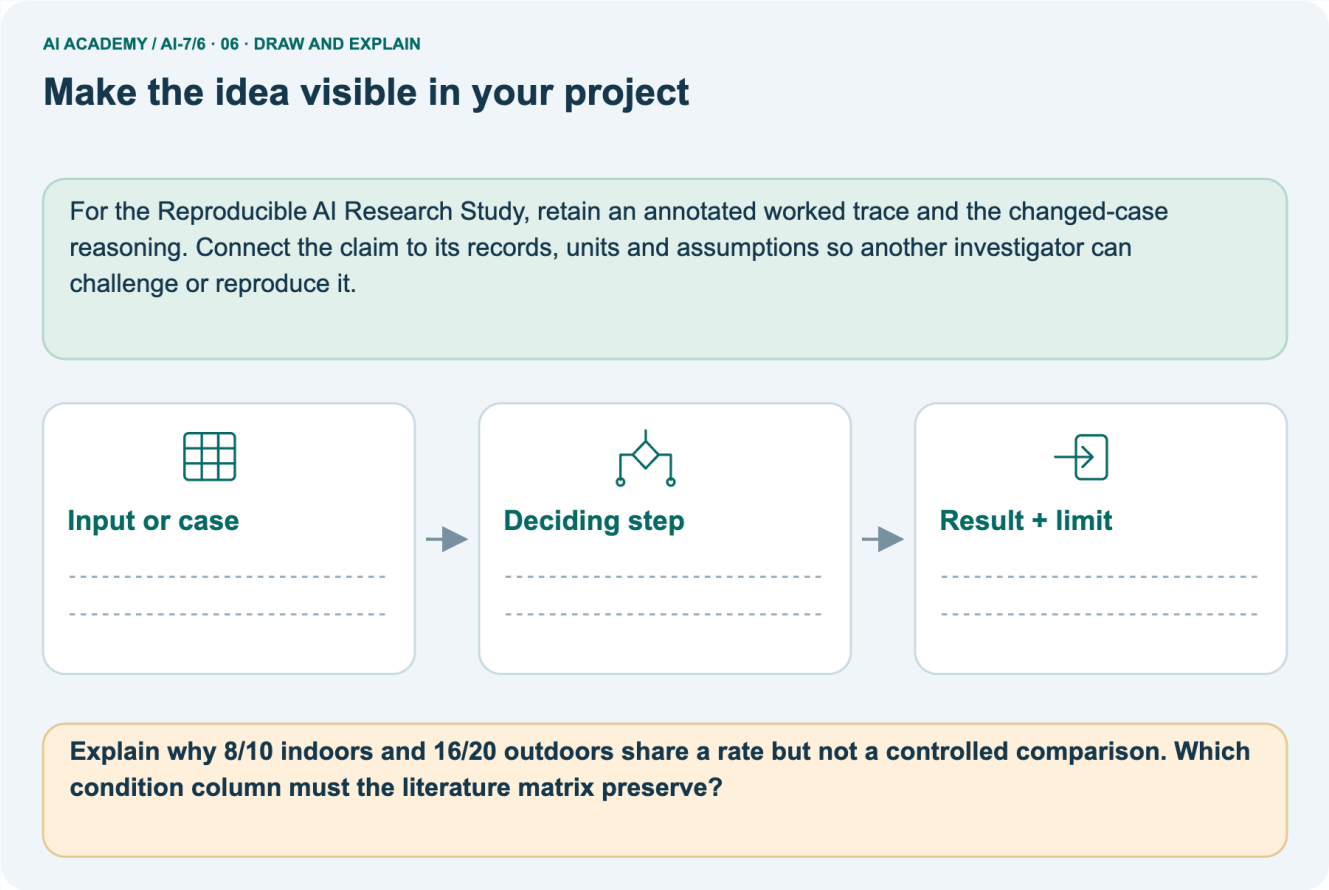
A **literature matrix** helps compare **evidence** without flattening differences. Equal percentages do not mean equivalent tasks, samples or methods.

Limit: These fictional studies illustrate comparison practice; no real literature finding is asserted.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the **literature matrix** preserve?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. A literature comparison matrix: the case

02. Follow the deciding steps

03. Same rate, different study conditions

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.